BB 101 BIOLOGY · END-SEMESTER EXAMINATION

Short Notes & Points to Remember

Everything in Lectures 15–23 on a few pages each: definitions, tables, diagrams, numbers, and the traps examiners like.

SYLLABUS
End-sem lecture slides, Lectures 15–23 (255 slides)

REFERENCE
Campbell Biology, 10th ed. — Ch. 11–15, 16.1, 21.2, 22–24

PREPARED FOR
Tutorial Batch T1

Contents

#	Topic	Lecture	Campbell
1	Heredity and Mendel’s laws	15–16	14.1
2	Mendel and the gene idea: probability, dominance, pedigrees	17	14.2–14.4
3	Cell cycle, mitosis and meiosis	18	12, 13
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5	Cell communication	20	11
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7	Evolution and natural selection	22	22, 23 (opening), 24.1
8	Population genetics and Hardy–Weinberg	22–23	23
9	Last-minute revision sheet: numbers, names, formulas, traps	all	—

HOW TO USE THESE NOTES

Read one section, then do the matching section (A–H) of the Practice Questions booklet. Teal boxes are must-remember facts; red boxes are common mistakes. Section 9 is for the night before the exam.

Written from the end-sem lecture slides, with explanations checked against Campbell Biology (10th ed.).

1 · Heredity and Mendel's laws

Lectures 15–16 · Campbell 14.1

The road to "DNA is the genetic material"

Year	Who	What
1866	Gregor Mendel	Rules of inheritance from thousands of pea crosses (7 characters, 3 : 1 in F_2)
1928	Frederick Griffith	'Transforming principle' — something from dead bacteria changes living ones
1944	Avery, MacLeod & McCarty	The transforming principle is DNA
1953	Watson & Crick	Double-helix structure of DNA

DNA is the genetic material of all known organisms (exception: some viruses use RNA). It is a self-driven programme that drives the development and controls the functioning of the organism.

Key terms

Term	Meaning
Character	A heritable feature that varies (flower colour).
Trait	One variant of a character (purple, white). Can be qualitative (eye colour) or quantitative (height, blood pressure).
Heritable trait	Phenotypic character passed from parents to offspring.
Phenotype / genotype	Observable property / genetic make-up. *PP* and *Pp*: same phenotype (purple), different genotypes.
Genome	An organism's full hereditary information (its DNA).
True-breeding	Self-pollination gives only the same variety (homozygous).
Hybridization	Crossing two contrasting true-breeding varieties.
P, F_1, F_2	Parents (true-breeding); hybrid offspring; offspring of F_1 self/cross-pollination. F = filial (child).
Gene / allele	Mendel's heritable 'unit' / 'factor' / an alternative version of a gene.

Why the garden pea? (model-system criteria)

- Available in many variations
- Short generation time
- Many offspring per mating
- Easy cross-pollination — pollen-producing and egg-bearing organs well separated
- Characters with distinct 'either-or' forms

The 7 characters (Campbell Table 14.1)

Flower colour, flower position, seed colour, seed shape, pod shape, pod colour, stem length.

The monohybrid cross

	Sperm *P* (½)	Sperm *p* (½)
Egg *P* (½)	*PP* purple	*Pp* purple
Egg *p* (½)	*Pp* purple	*pp* white

P: purple *PP* × white *pp* → F_1: all purple *Pp* (no blending) → F_2: 705 purple : 224 white ≈ **3 : 1**; genotypes **1 : 2 : 1**.

Mendel's two laws

Law	Statement	Physical basis (meiosis)
Segregation	The two alleles of **one gene** separate during gamete formation, so each gamete carries only one of them (½ *P*, ½ *p*).	Homologous chromosomes separate in anaphase I.
Independent assortment	Alleles of **different genes** sort into gametes independently — provided the genes are on different chromosomes or far apart on the same one. Dihybrid F_2: **9 : 3 : 3 : 1**.	Each homologous pair lines up at metaphase I independently of the other pairs.

REMEMBER

- The white factor was hidden in F_1, not destroyed — it came back unchanged in F_2. That disproves **blending** and supports the **particulate** hypothesis (discrete units).
- Mendel followed traits to at least F_2 and counted thousands of plants — quantitative analysis is what revealed the pattern.
- Independent assortment is why siblings (except identical twins) are never genetically the same.

TRAP

Segregation = **one** gene, two alleles. Independent assortment = **two or more** genes. A 'trait' is not the same as a 'character'.

2 · Mendel and the gene idea

Lecture 17 · Campbell 14.1–14.4

Vocabulary

- **Homozygous**: two identical alleles (*HH, zz, TT, aa*) — true-breeding.
- **Heterozygous**: two different alleles (*Jj, Bb, Cc*) — hybrid.
- **Locus**: a gene's position on a chromosome; alleles sit at the same locus on homologous chromosomes.
- **Punnett square**: all combinations of sperm and egg; capital = dominant, lowercase = recessive.
- **Testcross** (Campbell): unknown dominant × homozygous recessive. All dominant → *PP*; 1 : 1 → *Pp*.

Why is purple dominant?

The *P* allele codes for an **active enzyme** that makes purple pigment; a change in the DNA sequence of *p* gives an **inactive enzyme**. In *Pp*, **one functional allele suffices** to make enough pigment. (The protein need not be an enzyme.)

Probability rules

Rule	Use it when...	Example (*Rr* × *Rr*)
Multiplication	Two or more **independent** events must happen together ("and").	P(*rr*) = ½ × ½ = ¼
Addition	An outcome can happen in two or more **mutually exclusive** ways ("or").	P(heterozygote) = ¼ + ¼ = ½

METHOD FOR MULTI-GENE CROSSES

Split the cross into one monohybrid cross per gene, find each probability, then multiply. Slide problem: *PpYyRr* × *Ppyyrr*, P(recessive for at least two characters): *ppyyRr* 1/16 + *ppYyrr* 1/16 + *Ppyyrr* 2/16 + *PPyyrr* 1/16 + *ppyyrr* 1/16 = **6/16 = 3/8**.

Number of gamete types = 2^n (n = number of heterozygous genes): *AaBbCcDd* → 16.

Degrees of dominance and beyond

Pattern	What happens	Example
Complete dominance	Heterozygote looks like the dominant homozygote.	Purple vs white pea
Incomplete dominance	F_1 is intermediate between the parents; F_2 phenotype ratio = genotype ratio 1 : 2 : 1.	Snapdragons: red × white → pink
Codominance	Both alleles affect the phenotype in separate, distinguishable ways.	$I^A I^B$ → blood group AB
Multiple alleles	More than two alleles of a gene in the population.	ABO: I^A, I^B, i
Pleiotropy	One gene, many phenotypic effects.	Sickle-cell allele (Campbell)

Dominance depends on the level at which you look at the phenotype, and says nothing about how common an allele is (polydactyly is dominant but rare).

ABO blood groups

Genotype	Group	Carbohydrate on RBC
$I^A I^A$ or $I^A i$	A	A
$I^B I^B$ or $I^B i$	B	B
$I^A I^B$	AB	both
ii	O	neither

3 alleles → **6 genotypes, 4 phenotypes**. I^A and I^B are codominant; both are dominant over i.

Pedigrees — reading rules

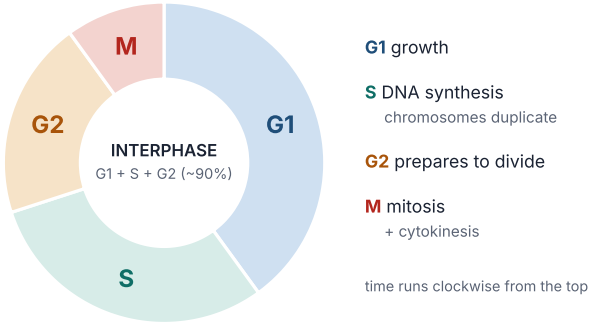
- □ male, ○ female, filled = affected; horizontal line = mating.
- Two **unaffected** parents with an **affected** child → **recessive** (both parents carriers, $Aa \times Aa$, risk ¼).
- Two **affected** parents with an **unaffected** child → **dominant**.
- Lecture examples: widow's peak = dominant; attached earlobe = recessive; albinism = recessive.
- Two carriers: P(affected) = ¼, but P(a **normal** child is a carrier) = **2/3**.
- Humans: long generations, few offspring, no breeding experiments → we use pedigrees.

3 · Cell cycle, mitosis and meiosis

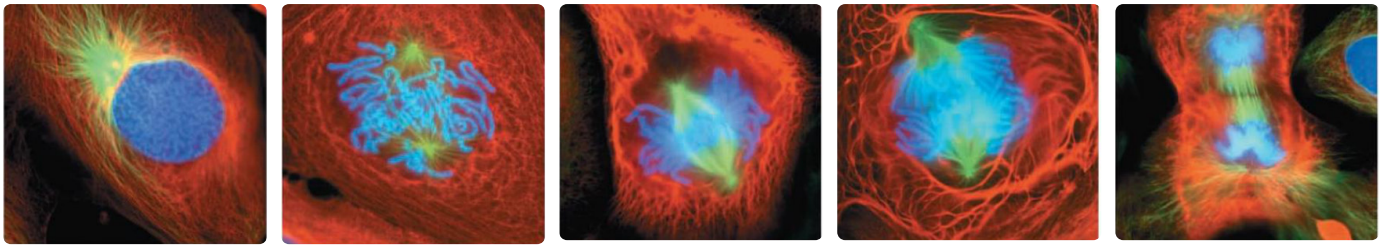
Lecture 18 · Campbell Ch. 12 and 13

Genetic material in the cell

- **Genome**: one DNA molecule (prokaryotes) or several (eukaryotes).
- **Chromatin**: DNA + protein; condenses into visible chromosomes during division.
- **Somatic cells**: two sets of chromosomes ($2n$ = 46 in humans; 23 pairs = 22 autosome pairs + XX/XY).
- **Gametes**: one set (n = 23).
- After S phase each chromosome = two **sister chromatids** joined at the **centromere** (the narrow 'waist'). Once separated, each chromatid is called a chromosome.
- Cell division lets cells proliferate — for growth, repair and reproduction.



Mitosis — the five stages



1 Prophase | 2 Prometaphase | 3 Metaphase | 4 Anaphase | 5 Telophase + cytokinesis

Lecture micrographs (Campbell Fig. 12.7): chromosomes blue, microtubules green, actin red.

Stage	What happens (Campbell 12.2)
Prophase	Chromatin condenses into chromosomes; nucleoli disappear; mitotic spindle begins to form; centrosomes move apart.
Prometaphase	**Nuclear envelope fragments**; kinetochores form at centromeres; some spindle microtubules attach to kinetochores and start moving chromosomes.
Metaphase	Chromosomes line up at the **metaphase plate** — an imaginary plane midway between the two poles.
Anaphase	Sister chromatids separate and move to opposite poles; kinetochore microtubules **shorten by depolymerizing at their kinetochore ends**; the cell elongates.
Telophase	Two daughter nuclei form; nuclear envelopes and nucleoli re-form; chromosomes decondense.
Cytokinesis	Animal cells: **cleavage furrow** (contractile ring of actin + myosin). Plant cells: **cell plate** (vesicles fuse in the middle and build a new wall).

Spindle = centrosomes + spindle microtubules + asters (radial arrays of short microtubules). The centrosome duplicates in interphase; the two centrosomes move to opposite ends during prophase and prometaphase. **Kinetochores** = protein complexes at the centromere.

Meiosis — one replication, two divisions

	Meiosis I (reduction division)	Meiosis II (like mitosis)
Prophase	**Synapsis** of homologues; **crossing over** between non-sister chromatids (chiasmata)	Spindle forms again
Metaphase	**Pairs** of homologues line up at the plate	Single chromosomes line up
Anaphase	**Homologues** separate; sister chromatids stay together	**Sister chromatids** separate
Result	2 haploid cells; each chromosome still has two chromatids	4 haploid cells with unduplicated chromosomes

Mitosis vs meiosis

Property	Mitosis	Meiosis
DNA replication	Once, in interphase before mitosis	Once, in interphase before meiosis I (none between I and II)
Number of divisions	One	Two
Synapsis & crossing over	Do not occur	Occur in prophase I
Daughter cells	Two diploid ($2n$), identical to the parent	Four haploid (n), different from the parent and from each other
Role	Growth, repair, asexual reproduction	Gametes; genetic diversity in sexual reproduction

Counting chromosomes and DNA

Stage (human)	Chromosomes	Chromatids	DNA
G_1	46	46	x
G_2 / metaphase	46	92	$2x$
After mitosis (each cell)	46	46	x
After meiosis I (each cell)	23	46	x
Gamete (after meiosis II)	23	23	$x/2$

Rule: count chromosomes by counting **centromeres**.

Why meiosis matters — sources of variation

- **Independent assortment**: 2^n combinations; humans $2^{23} \approx$ **8.4 million** per person.
- **Crossing over**: recombinant chromosomes with new allele combinations.
- **Random fertilization**: 8.4 million × 8.4 million ≈ **70 trillion** zygote combinations.
- Fertilization restores $2n$: the zygote carries a maternal and a paternal set.

TRAPS

- The nuclear envelope breaks up in **prometaphase**, not prophase.
- Mitosis divides the **nucleus**; cytokinesis divides the cytoplasm.
- Homologous chromosomes (one from each parent, same genes, maybe different alleles) ≠ sister chromatids (identical copies joined at the centromere).
- Anaphase I separates homologues; anaphase II (and mitotic anaphase) separates sister chromatids.

4 · Chromosomal basis of inheritance

Lecture 19 · Campbell Ch. 15

Chromosome theory of inheritance: Mendelian genes have specific loci on chromosomes, and it is the chromosomes that undergo segregation and independent assortment. Genes can be seen on isolated chromosomes by tagging them with a fluorescent dye.

Morgan's fruit flies (*Drosophila melanogaster*)

Cross	Result
P	Red-eyed (wild-type, w^+) female × white-eyed (mutant, w) male
F_1	All red-eyed → white is recessive
F_2	3 red : 1 white, but **all white-eyed flies were male**
Conclusion	The eye-colour gene is on the **X chromosome**, with no corresponding locus on the Y.

Sex chromosomes

- Mammals: larger X, smaller Y. Females XX, males XY.
- Only the ends of the Y have regions homologous with the X.
- **SRY** (Sex-determining Region of Y): its protein triggers male anatomical features (testes).
- Every ovum carries X; sperm are 50% X, 50% Y.
- The X carries many genes unrelated to sex; the Y mainly codes for sex determination. Other animals use other systems.

X-linked recessive inheritance

- **Sex-linked gene**: on a sex chromosome (mostly X-linked; few Y-linked).
- Female affected only if homozygous (X^nX^n); male affected with one copy — he is **hemizygous**.
- So X-linked recessive disorders are much more common in males.
- Examples: **colour blindness** (mostly X-linked), **Duchenne muscular dystrophy**, **haemophilia**.

Cross	Daughters	Sons
Colour-blind father × normal (homozygous) mother $X^nY \times X^NX^N$	All carriers, normal vision	All normal — father gives them his Y
Carrier mother × normal father $X^NX^n \times X^NY$	½ carriers, ½ normal	½ affected, ½ normal
Carrier mother × colour-blind father $X^NX^n \times X^nY$	½ affected, ½ carriers	½ affected, ½ normal (free of the allele)

Errors in chromosome number

- **Nondisjunction**: homologues (meiosis I) or sister chromatids (meiosis II) fail to separate.
- In meiosis I → gametes $n+1$, $n+1$, $n-1$, $n-1$ (all abnormal).
- In meiosis II → $n+1$, $n-1$, n, n.
- **Aneuploidy**: abnormal number of one chromosome. Trisomic = $2n+1$; monosomic = $2n-1$.
- **Polyploidy**: more than two complete sets (triploid $3n$, tetraploid $4n$). Common in plants, rare in animals; polyploids look more normal than aneuploids.

Condition	Karyotype	Note
Down syndrome	Trisomy 21	Frequency rises with mother's age
Klinefelter	XXY	Male
Turner	X0	Female, sterile; only known viable human monosomy

> **REMEMBER**
> A father passes his X to **every daughter** and his Y to **every son**. So a son's X-linked trait always comes from his mother.

5 · Cell communication

Lecture 20 · Campbell Ch. 11

Why it matters: fight-or-flight. Seeing a predator → adrenal glands secrete **adrenaline** (epinephrine) → muscles work harder, heart beats faster, breathing speeds up. Cells must communicate and coordinate.

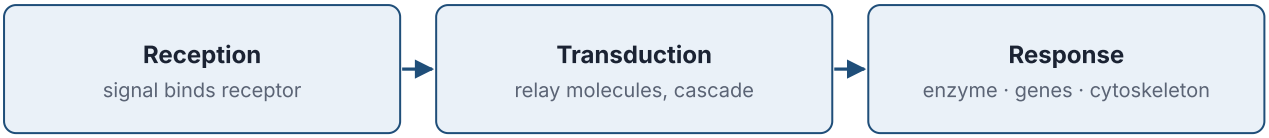
Signalling is ancient and universal

- *Myxococcus xanthus* (soil bacterium): starving cells send a chemical signal → aggregate → **fruiting bodies** with spores.
- **Quorum sensing**: bacteria sense local population density from the concentration of signalling molecules.
- **Yeast mating** (*Saccharomyces cerevisiae*): type a secretes **a factor** → binds receptors on α cells; α secretes α factor → receptors on a cells → cells fuse → **a/α diploid** (can do mitosis or meiosis).
- Mechanisms are evolutionarily conserved — the same ones recur from microbes to humans.

Local vs long-distance signalling

Type	How
Cell junctions	Direct cytoplasm-to-cytoplasm: **gap junctions** (animals), **plasmodesmata** (plants)
Cell–cell recognition	Membrane-bound surface molecules; embryonic development, immune response
Paracrine	Local regulators (e.g. **growth factors**) act on nearby cells
Synaptic	Electrical signal → **neurotransmitter** across synapse → electrical signal
Endocrine (long-distance)	**Hormones** travel in blood; act only on cells with receptors

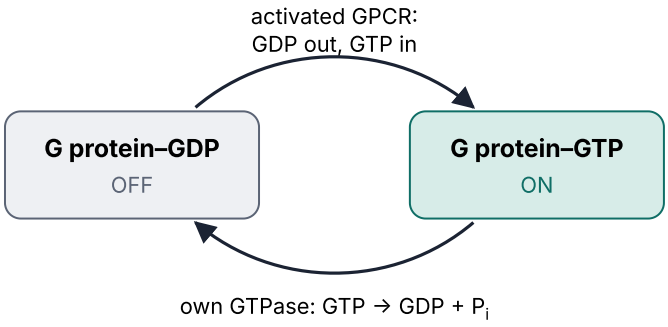
Three stages: reception → transduction → response



Doorbell analogy: knock received (discriminator) → message carried inside (transducer) → action triggered (amplifier)

Transduce = to convert energy or a message into another form. Binding changes the receptor's **shape**; the shape change is passed on, often through several relay molecules. Responses: (1) catalysis by an enzyme, (2) activation of specific genes, (3) rearrangement of the cytoskeleton.

G proteins — molecular switches



- Large, ubiquitous family; ATP = energy currency, GTP = used in signal transduction.
- Built-in GTPase = built-in control: the response is **transient**.
- Nobel 1994: **Alfred Gilman & Martin Rodbell** — discovery of G proteins.

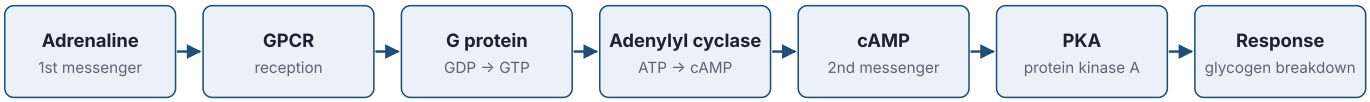
GPCRs — G protein-coupled receptors

- **Seven transmembrane domains**; extremely widespread (development, vision, taste, smell…).
- **~60% of all medicines** target GPCRs (β-blockers, albuterol, antipsychotics…).
- Involved in cholera, pertussis, botulism. Cholera toxin locks a G protein ON → cAMP stays high → salt and water secretion (Campbell 11.3).

Activation steps

1. Ligand binds the GPCR outside the cell.
2. Receptor changes shape.
3. Inactive G protein (GDP) binds inside.
4. GDP swapped for GTP → G protein active, leaves the receptor, activates an enzyme.
5. GTP hydrolysed → G protein off; ligand binding is reversible.

The adrenaline → cAMP pathway (Sutherland, Nobel 1971)



Phosphodiesterase switches the signal off: cAMP → AMP

Amplification (Campbell Fig. 11.16)

Step	Molecules
Adrenaline bound to GPCR (reception)	1
Active G proteins	10^2
Active adenylyl cyclase	10^2
Cyclic AMP	10^4
Active protein kinase A	10^4
Active phosphorylase kinase	10^5
Active glycogen phosphorylase	10^6
Glucose 1-phosphate released (response)	10^8

ADVANTAGE OF A CASCADE (SLIDE: "WHAT IS THE ADVANTAGE?")

- **Amplification** — one hormone molecule → ~10^8 product molecules.
- **Regulation** — many points where the signal can be tuned or stopped.
- **Coordination / branching** — one signal can trigger several responses.

TRAPS

- G protein ON = **GTP**; OFF = GDP.
- **Adenylyl cyclase** makes cAMP from ATP; **phosphodiesterase** destroys it (→ AMP).
- Adrenaline never enters the cell — it is the **first** messenger; cAMP is the second.

Engineering link from the lecture: signalling is a feedback-and-control system (Systems & Control Engineering, IIT Bombay) — sensor (receptor), transducer, amplifier, actuator, and a built-in switch-off.

6 · Big data in biology and bioinformatics

Lecture 21 · slides; Campbell 16.1, 21.2

-ome and -omics

Suffix	Meaning
-ome	The complete set
-omics	The study of an -ome

-ome	Complete set of…
Genome	DNA (hereditary information)
Transcriptome	RNA transcripts
Proteome	Proteins
Metabolome	Small-molecule metabolites
Lipidome / Glycome	Lipids / sugars (glycans)
Epigenome	Epigenetic marks
Microbiome	Microbial community and its genes

-Omics are **complementary** to one-protein-at-a-time studies: different information, different questions.

Why 'big data'?

- High-throughput -omics; data generation has become **affordable**.
- Advances in many fields: chemistry (ddNTPs, dyes), physics (optics), engineering (automation), computer science (algorithms).
- Genomics (PLoS Biol 2015): **~1 zetta-base** acquired per year, **2–40 exabytes** of storage — compared with astronomy, Twitter, YouTube.
- Data types: images/video, text, numbers, audio (even plants 'cry' when stressed).
- Same-day whole-genome sequencing is now being tried in critical care (NEJM, 2025).

Prefix	Power	Prefix	Power
kilo	10^3	peta	10^{15}
mega	10^6	exa	10^{18}
giga	10^9	zetta	10^{21}
tera	10^{12}		

DNA double helix: A, B and Z forms

Parameter	A-DNA	B-DNA (common)	Z-DNA
Handedness	right	right	**left**
Base pairs per turn	11	**10**	12
Rotation per base pair	33°	36°	−30°
Rise per base pair	2.55 Å	**3.4 Å**	3.7 Å
Helix diameter	23 Å	**20 Å**	18 Å

B-DNA: one turn = 10 × 3.4 Å = 3.4 nm; width 2 nm (Campbell 16.1). Human haploid genome ≈ 3×10^9 bp × 0.34 nm ≈ 1 m; a diploid cell holds ≈ 2 m of DNA.

Bioinformatics

- **Broad definition**: analyse biological data to extract patterns and generate hypotheses (blind men and the elephant: each discipline sees only a part).
- **Beginning (~1960)**: Margaret Dayhoff, Richard Eck, Robert Ledley — computer-aided analysis of protein sequences → **Protein Information Resource (PIR)**.
- **Higgs & Attwood**: (1) computational methods for studying the structure, function and evolution of genes, proteins and genomes; (2) methods to manage and analyse information from genomics and high-throughput experiments.

Three fields meet

Field	Contributes
Statistics	Data analysis, choice of metrics, pattern extraction, validity of inferences
Computer science	Algorithms; storage and retrieval (query optimisation, transmission, archiving)
Biology (domain knowledge)	Interpretation, hypothesis testing, context for data generation; the why, how and what; ground-truth data

Why compare sequences?

- Infer function from a better-characterised sequence
- Find evolutionary relationships

- **Goals**: predict biological processes in health and disease; integrate gene/protein information; build and improve tools; better data management, search and retrieval. Success ultimately depends on our knowledge of biology.

AI/ML: the black box

Big data → AI/ML algorithm → prediction. A **black box** is a system whose internal workings are hidden. Biologists ask: *why* this prediction, and what is the underlying biological pattern?

- **Explainable methods**: post hoc understanding of what a trained model has learnt.
- **Interpretable models**: intelligible by design (parameters and architecture are understandable).

- Find similar regions (catalytic or binding sites)

Applications

- Database similarity search; structure prediction; annotation
- Assembling reads into genomes
- Mapping reads to a reference ('resequencing'): SNPs, indels
- ChIP-Seq: transcription-factor binding sites

Protein structure

- Atomic detail: **X-ray crystallography**, **NMR**, **electron microscopy** (~1 Å upward). AFM covers nm-scale; **light microscopy** works at cell scale and cannot see atoms.
- **AlphaFold**: amino-acid sequence + evolutionary history + known 3D structures → deep neural networks → atomic-level 3D structure; very high, independently verified accuracy; huge savings in time, people and cost.
- **Nobel Chemistry 2024**: David Baker, Demis Hassabis, John Jumper.

7 · Evolution and natural selection

Lecture 22 · Campbell Ch. 22, 23 (opening), 24.1

Evolution: the process of change that has transformed life on Earth from its early beginnings to its present diversity. It underpins work on virus evolution, cancer progression and drug resistance.

Darwin

- Before Darwin: species were seen as created by God and unchanging.
- Voyage on **HMS Beagle**; specimens from all over the world, especially islands (isolated populations) — the **Galápagos** (finches, tortoises, iguanas).
- *On the Origin of Species by Means of Natural Selection*, **1859**.
- Three observations: the **unity** of life, the **diversity** of life, and the **match** between organisms and their environments.
- Today's organisms descend from ancestral species (descent with modification); the mechanism is **natural selection**.

Biological species concept

A group of populations whose members can **interbreed in nature** and produce **viable, fertile offspring** — but not with members of other such groups.

- Male donkey × female horse = **mule**; female donkey × male horse = **hinny**. Both sterile → separate species.
- Organisms that only divide to multiply (asexual) are excluded.

Galápagos finches

Finch	Beak	Food
Cactus ground finch	Long, sharp	Cactus flowers, pulp
Green warbler finch	Narrow, pointed	Insects
Large ground finch	Large, short	Seeds from the ground

Adaptation: inherited characteristics that enhance survival and reproduction in specific environments.

Darwin's observations and inferences (Concept 22.2)

Observations	Inferences
#1 Members of a population vary in their inherited traits (e.g. ladybird beetle colour/camouflage).	#1 Individuals whose traits give a higher chance of surviving and reproducing leave more offspring.
#2 All species produce more offspring than the environment can support; many fail to survive (puffball spore cloud). Malthus: populations grow exponentially, food linearly.	#2 This unequal ability to survive and reproduce leads to the accumulation of favourable traits in the population over generations.

The four components: **variation**, **inheritance**, **high rate of population growth**, **differential survival and reproduction**. Source of variation: DNA polymerase errs ~1 in 10^4–10^5 nucleotides; 10^5–10^6 errors for a diploid mammalian cell; most are repaired, some remain.

Evolution observed: the medium ground finch

- Daphne Major, Galápagos; drought of 1977.
- Small, soft seeds scarce; large, hard seeds plentiful.
- ~1200 birds → ~180 survived (deep-beaked birds could crack hard seeds).
- Average beak depth rose in the next generation (≈ 9 → ≈ 9.7 mm on the slide's graph).
- **Individual finches did NOT evolve** — no beak grew. The **proportion** of large-beaked birds rose: the population evolved.

Evolution observed: MRSA

- *Staphylococcus aureus*: penicillin resistance 1945; methicillin resistance 1961 — each about two years after wide use.
- Methicillin inhibits a protein bacteria use to build cell walls; MRSA uses a different protein, survives and out-reproduces the others.
- The drug does not cause the mutation — it selects for it. (Also: Harvard's 'mega-plate' Petri dish experiment.)

THREE KEY POINTS ABOUT NATURAL SELECTION

1. Selection acts on individuals, but **populations evolve**.
2. It can only amplify or diminish **heritable** traits that already **differ** among individuals.
3. Environments vary in place and time, so a trait that helps in one place or time may be useless or harmful in another.

Natural selection	Artificial selection
Nature 'decides'; works on individuals; e.g. finch beaks	Humans decide; selective breeding; inbreeding occurs; e.g. Dalmatians

Darwin's argument: if humans can cause big changes in a short time, nature can do so over a long time.

8 · Population genetics and Hardy–Weinberg

Lectures 22–23 · Campbell Ch. 23

Darwin could not explain how variation is maintained; Mendel's discrete genes (not blending) provided the answer. Population genetics is the foundation for studying evolution.

The Hardy–Weinberg theorem

Allele and genotype frequencies in a gene pool **remain constant** from generation to generation unless acted on by agents other than sexual recombination (gene shuffling in meiosis). Such a population is **not evolving**.

FORMULAS (ONE GENE, TWO ALLELES A AND A)

p = freq(A), q = freq(a) $\quad \boldsymbol{p + q = 1}$

$\boldsymbol{p^2 + 2pq + q^2 = 1}$ → AA, Aa, aa

Five conditions for equilibrium

Condition	If violated →
Extremely large population	Genetic drift
No gene flow	Gene flow
No mutations	Mutation
Random mating	Sexual selection / inbreeding
No natural selection	Natural selection

Microevolution = generation-to-generation change in a population's allele frequencies.

SOLVING A HARDY–WEINBERG PROBLEM

1. Find q^2 = fraction showing the **recessive** phenotype (the only class you can read directly).
2. $q = \sqrt{(q^2)}$; $p = 1 - q$.
3. Heterozygotes = $2pq$; homozygous dominant = p^2; multiply by N for numbers.
4. If genotype counts are given, **count alleles** instead: $p = (2 \cdot AA + Aa)/(2N)$. Compare observed with $p^2 : 2pq : q^2$ to test equilibrium.

Worked example	Working	Answer
Slide: $q^2 = 0.09$	$q = 0.3$, $p = 0.7$; $p^2 = 0.49$, $2pq = 0.42$	0.49 : 0.42 : 0.09
Slide: 552 plants, 546 purple	$q^2 = 6/552 = 0.0109 \rightarrow q = 0.104$	$q \approx 0.1$, $p \approx 0.9$
Slide: lizards 1024 *GG*, 512 *Gg*, 64 *gg*	$G = 2\times1024 + 512 = 2560$; $g = 512 + 2\times64 = 640$; total 3200	$p = 0.8$, $q = 0.2$; 0.64 : 0.32 : 0.04
Campbell: PKU, 1 in 10,000	$q = 0.01$, $p = 0.99$; $2pq = 0.0198$	≈ 2% carriers

Causes of microevolution (lecture list)

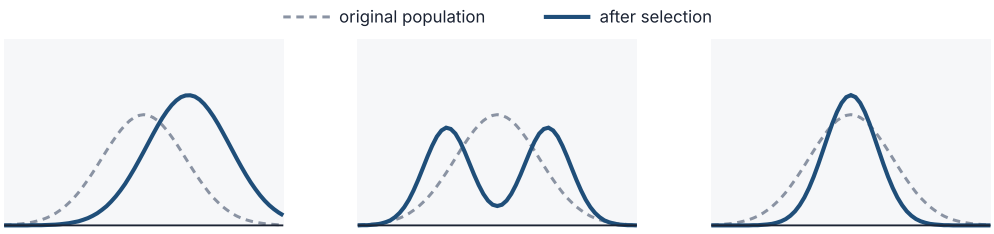
Cause	What it does
Mutation	Changes in DNA (point mutations, gene duplication) alter or create new alleles.
Sexual recombination	Rearranges alleles into fresh combinations every generation.
Natural selection	Individuals with certain heritable traits survive and reproduce at higher rates.
Genetic drift	Chance changes in allele frequencies; strongest in small populations. **Bottleneck effect** (drastic reduction in size) and **founder effect** (a few individuals colonise a new area). Slide example: $p = 0.7 \rightarrow 0.5 \rightarrow 1.0$ (allele *r* lost by chance).
Gene flow	Movement of alleles by migration of fertile individuals or gametes (wind-blown pollen); reduces differences between populations.

SLIDES VS CAMPBELL

The slides list sexual recombination as a cause; Campbell 10th ed. names **natural selection, genetic drift and gene flow** as the three mechanisms that directly change allele frequencies, with mutation and recombination as sources of variation. Recombination makes new combinations but, by itself, does not change allele frequencies (that is what Hardy–Weinberg says).

Fitness and modes of selection

Fitness: the contribution an individual makes to the gene pool of the next generation (relative to others). Natural selection alters the frequency distribution of heritable traits in three ways:



Directional
one extreme favoured; mean shifts

Disruptive (diversifying)
both extremes favoured

Stabilizing
intermediates favoured; variation shrinks

Mode	Example on the slide
Directional	Beak sizes of finches during wet/dry seasons in the Galápagos
Disruptive (diversifying)	Small beaks for small seeds; large beaks for large seeds
Stabilizing	Average human birth weight

9 · Last-minute revision sheet

Numbers, names, formulas, ratios and traps on two pages

Numbers

Human chromosomes	$2n = 46$, $n = 23$; 92 chromatids in G_2
Independent assortment (humans)	$2^{23} \approx 8.4$ million; × fertilization ≈ 70 trillion
Mendel's F_2	705 : 224 ≈ 3.15 : 1
ABO	3 alleles, 6 genotypes, 4 phenotypes
Interphase	~90% of the cell cycle
B-DNA	10 bp/turn, 3.4 Å/bp, 20 Å wide, right-handed
Z-DNA	12 bp/turn, left-handed; A-DNA 11 bp/turn
GPCR	7 transmembrane domains; ~60% of drugs
Adrenaline cascade	1 → 10^8 glucose 1-phosphate
Genomics data (2015)	~1 zetta-base/yr; 2–40 exabytes stored
DNA polymerase error rate	~1 in 10^4–10^5 nucleotides
Daphne Major drought (1977)	~1200 → ~180 finches

Ratios

Monohybrid F_2	3 : 1 (phenotype), 1 : 2 : 1 (genotype)
Dihybrid F_2	9 : 3 : 3 : 1
Testcross $Pp \times pp$	1 : 1
Incomplete dominance F_2	1 red : 2 pink : 1 white
Morgan F_2	3 red : 1 white (all white = male)
Trihybrid slide problem	3/8

Names and dates

Mendel	1866 — laws of inheritance
Darwin	1859 — *Origin of Species*; HMS Beagle
Griffith	1928 — transforming principle
Avery, MacLeod, McCarty	1944 — DNA is the transforming principle
Watson & Crick	1953 — double helix
Morgan	X-linked white-eye gene in *Drosophila*
Earl Sutherland	Nobel 1971 — hormone action, cAMP
Gilman & Rodbell	Nobel 1994 — G proteins
Dayhoff, Eck, Ledley	~1960 — bioinformatics; PIR
Baker, Hassabis, Jumper	Nobel Chemistry 2024 — protein design, AlphaFold
Malthus	Populations grow exponentially, food linearly
Peter & Rosemary Grant (Campbell)	Medium ground finch, Daphne Major

Formulas

Gamete types	2^n (n heterozygous genes)
"and" / "or"	multiply / add
Hardy–Weinberg	$p + q = 1$; $p^2 + 2pq + q^2 = 1$
Allele counting	p = (2·AA + Aa) / 2N
DNA length	bp × 0.34 nm (B-DNA)

TWELVE CLASSIC TRAPS

1. Character (flower colour) ≠ trait (purple).
2. Segregation = one gene; independent assortment = different genes.
3. P(normal child of two carriers is a carrier) = 2/3, not 1/2.
4. Codominance (AB) ≠ incomplete dominance (pink).
5. Nuclear envelope breaks up in prometaphase.
6. No DNA replication between meiosis I and II.
7. Anaphase I: homologues separate; anaphase II: sister chromatids.
8. Sons get their X from their mother — never X-linked traits from their father.
9. G protein: GTP = ON; adenylyl cyclase makes cAMP; phosphodiesterase breaks it.
10. -Omics complement, not replace, single-molecule studies.
11. Individuals don't evolve; populations do. Selection only acts on existing heritable variation.
12. Hardy–Weinberg: start from the recessive phenotype (q^2), never from the dominant one.